Appendix A
Selected Derivations

The technical notes are a companion track to this book, not a part of it. They hold the derivations and the technical detail stripped out of the main text so that the chapters can stay intuitive, they are maintained and published separately, and the chapters point into them with markers such as [TN14].

This appendix reproduces the two the main argument leans on hardest, so that a reader working from the book alone is not stopped by them: the quadratic-variation and Itô arguments behind Chapter 14, and the integral behind Chapter 15's formula. Each keeps its own equation prefix, so a reference to (TN15.4) means the same thing here as it does in the companion track.

A.1 ITÔ'S LEMMA: DERIVATIONS

This note supplies the arguments Chapter 14 states without proof: why $dW^2 = dt$, how Itô's lemma follows, and why a change of measure alters the drift of a diffusion but not its volatility. The level is that of a careful heuristic rather than a measure-theoretic treatment; readers wanting the latter should consult the references at the end.

A.1.1 The Quadratic Variation of a Wiener Process

Everything rests on one fact: over a fixed interval, the sum of *squared* increments of a Wiener process converges to the length of the interval, and does so with vanishing randomness.

Partition $[0, T]$ into n intervals of length $\Delta t = T/n$, and write ΔW_i for the increment over the ith. From the definition, $\Delta W_i = \varepsilon_i \sqrt{\Delta t}$ with ε_i independent standard normals. Consider

$$Q_n = \sum_{i=1}^{n} \Delta W_i^2 = \Delta t \sum_{i=1}^{n} \varepsilon_i^2 \tag{TN14.1}$$

The mean. Since $\mathbb{E}[\varepsilon_i^2] = 1$,

$$\mathbb{E}[Q_n] = \Delta t \cdot n = T \tag{TN14.2}$$

independently of n.

The variance. For a standard normal, $\mathbb{E}[\varepsilon^4] = 3$, so $\mathrm{Var}(\varepsilon_i^2) = 3 - 1 = 2$. The ε_i are independent, so

$$\mathrm{Var}(Q_n) = \Delta t^2 \cdot n \cdot 2 = 2\,\frac{T^2}{n^2} \cdot n = \frac{2T^2}{n} \tag{TN14.3}$$

As $n \to \infty$ the variance vanishes while the mean stays at T. Therefore

$$\sum_{i=1}^{n} \Delta W_i^2 \;\longrightarrow\; T \quad \text{in mean square} \tag{TN14.4}$$

The limit is not a random variable. This is the content of the informal statement $dW^2 = dt$: squared increments accumulate deterministically, at rate one per unit time.

It is worth contrasting this with the increments themselves. By the same partition, $\sum_i \Delta W_i = W(T) - W(0) \sim N(0, T)$, which remains random however fine the partition. Squaring destroys the randomness; adding does not.

Why ordinary calculus fails. Consider instead a smooth function $x(t)$ with bounded derivative. Then $\Delta x_i = O(\Delta t)$, so $\sum_i \Delta x_i^2 = O(n\,\Delta t^2) = O(\Delta t) \to 0$. For smooth functions the quadratic variation is *zero*, which is precisely why second-order terms are discarded in ordinary Taylor expansions. A Wiener path has $\Delta W_i = O(\sqrt{\Delta t})$, so its quadratic variation is $O(n\,\Delta t) = O(T)$—finite and nonzero. *The whole difference between the two calculi is the difference between $O(\Delta t)$ and $O(\sqrt{\Delta t})$ increments.*

Two corollaries follow by the same argument and are used below:

$$\sum_i \Delta W_i\,\Delta t \to 0, \qquad \sum_i \Delta t^2 \to 0 \tag{TN14.5}$$

so in the shorthand of Chapter 14, $dW\,dt = 0$ and $dt^2 = 0$, while $dW^2 = dt$ alone survives.

A.1.2 Itô's Lemma

Let S follow the Itô process

$$dS = a(S,t)\,dt + b(S,t)\,dW \tag{TN14.6}$$

and let $f(S,t)$ be twice differentiable in S and once in t. Expand f to second order:

$$\Delta f = \frac{\partial f}{\partial S}\Delta S + \frac{\partial f}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\Delta S^2 + \frac{\partial^2 f}{\partial S\,\partial t}\Delta S\,\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial t^2}\Delta t^2 + \cdots \tag{TN14.7}$$

Now substitute the discrete form of (TN14.6), namely $\Delta S = a\,\Delta t + b\,\varepsilon\sqrt{\Delta t}$, and count orders.

The squared term is

$$\Delta S^2 = a^2\Delta t^2 + 2ab\,\varepsilon\,\Delta t^{3/2} + b^2\varepsilon^2\,\Delta t \tag{TN14.8}$$

Of these three, the first two are $o(\Delta t)$ and vanish when summed, by (TN14.5). The third is $O(\Delta t)$ and, by Section A.1.1, its sum converges to $\int b^2\,dt$ with no residual randomness. Hence

$$\Delta S^2 \;\longrightarrow\; b^2\,dt \tag{TN14.9}$$

The remaining second-order terms in (TN14.7) contain $\Delta S\,\Delta t = O(\Delta t^{3/2})$ and Δt^2, both of which vanish. Substituting (TN14.9) and (TN14.6) into (TN14.7) and passing to the limit gives **Itô's lemma**:

$$\boxed{df = \left(\frac{\partial f}{\partial S}a + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}b^2\right)dt + \frac{\partial f}{\partial S}\,b\,dW} \tag{TN14.10}$$

Setting $a = \mu S$ and $b = \sigma S$ recovers equation (14.8) of the main text.

The two-variable case. For $f(x,y,t)$ with $dx = a_1 dt + b_1 dW_1$, $dy = a_2 dt + b_2 dW_2$ and $dW_1 dW_2 = \rho\,dt$, the same expansion carries an additional cross term $f_{xy}\Delta x\Delta y$. By the argument of Section A.1.1 applied to the product,

$$\sum_i \Delta x_i\,\Delta y_i \;\longrightarrow\; \int b_1 b_2 \rho\,dt \tag{TN14.11}$$

which produces the extra drift term $f_{xy}b_1b_2\rho$ in equation (14.18).

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A.1.3 The Lognormal Solution

Chapter 14 asserts that geometric Brownian motion has the explicit solution used for simulation. Apply (TN14.10) to $G = \ln S$ with $a = \mu S$ and $b = \sigma S$:

$$\frac{\partial G}{\partial S} = \frac{1}{S}, \qquad \frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}, \qquad \frac{\partial G}{\partial t} = 0$$

so

$$d \ln S = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma \, dW \tag{TN14.12}$$

The coefficients are constant, so this integrates directly:

$$\ln S_T = \ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma \, (W_T - W_0) \tag{TN14.13}$$

and since $W_T - W_0 \sim N(0, T)$,

$$\ln S_T \sim N\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T, \quad \sigma^2 T\right) \tag{TN14.14}$$

Exponentiating (TN14.13) gives the exact simulation scheme of equation (14.6).

Moments. If $\ln X \sim N(m, s^2)$ then, completing the square in the defining integral,

$$\mathbb{E}\left[X^k\right] = \mathbb{E}\left[e^{k \ln X}\right] = e^{km + k^2 s^2/2} \tag{TN14.15}$$

Setting $k = 1$ gives $\mathbb{E}[X] = e^{m + s^2/2}$; setting $k = 2$ and subtracting the square of the mean gives the variance quoted in the main text. Applying (TN14.15) with $k = 1$ to (TN14.14),

$$\mathbb{E}[S_T] = \exp\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T + \frac{\sigma^2 T}{2}\right) = S_0 e^{\mu T}$$

confirming that the Itô correction is exactly what makes the expected price grow at μ.

A.1.4 Why the Measure Change Preserves Volatility

Chapter 14 states that moving to the risk-neutral measure changes a diffusion's drift but leaves its volatility untouched. The reason is visible from Section A.1.1 without any formal apparatus.

A change of probability measure re-weights how likely each path is. It does not alter the paths themselves, and volatility is a property of a path rather than of its probability: by (TN14.4), the quadratic variation of S along any given path is $\int \sigma^2 S^2 \, dt$, computed from the path alone. Since two equivalent measures agree on which paths are possible—they differ only in the weights—they must agree on quadratic variation, and therefore on σ.

The drift enjoys no such protection. It is defined by an *expectation*, and expectations are exactly what re-weighting changes.

The formal statement. Girsanov's theorem makes this precise. If W is a Wiener process under P and λ is a suitably regular process, then

$$W_t^Q = W_t + \int_0^t \lambda_u \, du \tag{TN14.16}$$

is a Wiener process under a measure Q equivalent to P, with the change of measure given by the density

$$\frac{dQ}{dP} = \exp\left(-\int_0^T \lambda_u \, dW_u - \frac{1}{2}\int_0^T \lambda_u^2 \, du\right) \tag{TN14.17}$$

Substituting (TN14.16) into $dS = \mu S \, dt + \sigma S \, dW$ gives

$$dS = (\mu - \lambda\sigma) S \, dt + \sigma S \, dW^Q$$

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so choosing $\lambda = (\mu - r)/\sigma$—the **market price of risk**—produces the risk-neutral dynamics $dS = rS\,dt + \sigma S\,dW^{Q}$. The volatility is unchanged, as claimed, and only the drift has moved.

The quantity λ deserves a name because it recurs: it is the excess return per unit of volatility, and Chapter 28 shows that changing numeraire amounts to changing λ, which is why the convexity, timing, and quanto adjustments of Chapter 30 all take the form of a drift correction and never a volatility correction.

Further Reading

Neftci's *An Introduction to the Mathematics of Financial Derivatives* works through the arguments above at greater length and with more attention to what is being glossed over. Baxter and Rennie's *Financial Calculus* gives a short account of Girsanov's theorem and the martingale representation theorem, and is the place to start on both. Readers wanting the measure-theoretic treatment should go to Karatzas and Shreve, or to Shreve's second volume, which is written for exactly this audience.

A.2 THE BLACK–SCHOLES–MERTON FORMULA: DERIVATIONS

Chapter 15 states the formula and explains what its terms mean. This note carries out the integral that produces it, shows the equivalence with solving the differential equation, and derives the dividend and American-approximation results the chapter uses.

A.2.1 The Risk-Neutral Integral

Under the risk-neutral measure, Section A.1.3 established that

$$\ln S_T \sim N\!\left(m,\, s^2\right), \qquad m = \ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T, \qquad s = \sigma\sqrt{T} \tag{TN15.1}$$

and the call is the discounted expected payoff,

$$c = e^{-rT}\,\mathbb{E}^{Q}[\max(S_T - K,\,0)] = e^{-rT}\,\mathbb{E}^{Q}[(S_T - K)\mathbf{1}\{S_T > K\}] \tag{TN15.2}$$

Split it into two pieces:

$$c = e^{-rT}\underbrace{\mathbb{E}^{Q}[S_T\mathbf{1}\{S_T > K\}]}_{A} - e^{-rT}K\underbrace{Q(S_T > K)}_{B} \tag{TN15.3}$$

The second piece: B

Write $\ln S_T = m + sZ$ with Z standard normal. Then $S_T > K$ if and only if

$$Z > \frac{\ln K - m}{s}$$

so, using the symmetry $1 - N(x) = N(-x)$,

$$B = N\!\left(\frac{m - \ln K}{s}\right) = N\!\left(\frac{\ln(S_0/K) + \left(r - \sigma^2/2\right)T}{\sigma\sqrt{T}}\right) = N(d_2) \tag{TN15.4}$$

which identifies d_2 and confirms Chapter 15's statement that $N(d_2)$ is the risk-neutral probability of exercise.

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The first piece: A

Here the useful step is to complete the square. Writing $S_T = e^{m+sZ}$ and ϕ for the standard normal density,

$$A = \int_{z^*}^{\infty} e^{m+sz}\,\phi(z)\,dz = \frac{1}{\sqrt{2\pi}}\int_{z^*}^{\infty} \exp\!\left(m + sz - \frac{z^2}{2}\right) dz, \qquad z^* = \frac{\ln K - m}{s}$$

The exponent is

$$m + sz - \frac{z^2}{2} = m + \frac{s^2}{2} - \frac{(z-s)^2}{2}$$

so, substituting $y = z - s$,

$$A = e^{m+s^2/2}\,\frac{1}{\sqrt{2\pi}}\int_{z^*-s}^{\infty} e^{-y^2/2}\,dy = e^{m+s^2/2}\,N(s - z^*) \tag{TN15.5}$$

Two substitutions finish it. First, $e^{m+s^2/2} = S_0 e^{rT}$, the lognormal moment result (TN14.15) and the same cancellation of $\sigma^2/2$ noted there. Second,

$$s - z^* = \frac{m - \ln K}{s} + s = d_2 + \sigma\sqrt{T} = d_1$$

so $A = S_0 e^{rT} N(d_1)$.

Assembling

Substituting A and B into (TN15.3):

$$\boxed{c = S_0 N(d_1) - K e^{-rT} N(d_2)} \tag{TN15.6}$$

The put follows from put–call parity rather than a second integral, since parity holds identically.

$N(d_2)$ arose as a probability; $N(d_1)$ arose from completing the square, that is, from the shift $z \mapsto z - s$ induced by weighting each state by S_T. That shift is the content of Chapter 15's remark that $N(d_1)$ is not a risk-neutral probability: it is the probability of exercise computed under the measure in which the stock, rather than cash, is the numeraire. Chapter 28 develops that idea in general.

A.2.2 Equivalence with the Differential Equation

Chapter 15 offers two routes to (TN15.6). They agree, and here is why.

Define $V(S,t) = e^{-r(T-t)}\mathbb{E}_t^{Q}[h(S_T)]$ for any payoff h. By the law of iterated expectations, the discounted value $e^{-rt}V$ is a Q-martingale. Applying Itô's lemma to $e^{-rt}V(S,t)$ under the risk-neutral dynamics $dS = rS\,dt + \sigma S\,dW^{Q}$:

$$d\big(e^{-rt}V\big) = e^{-rt}\left(V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV\right)dt + e^{-rt}\sigma S V_S\,dW^{Q}$$

A martingale has zero drift. Setting the dt coefficient to zero gives

$$V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV \tag{TN15.7}$$

which is equation (15.1). *The differential equation and the risk-neutral expectation are the same statement*: one says the discounted price has no drift, the other computes the expectation that follows. This equivalence is the Feynman–Kac relation, and it is why Chapter 20 can solve pricing problems either by simulating expectations or by solving partial differential equations.

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A.2.3 Dividends

A continuous yield. With a yield q, the risk-neutral drift becomes $r - q$: the asset's total return must be r, and q of it arrives as income, so the price grows at $r - q$. Repeating Section A.2.1 with $m = \ln S_0 + (r - q - \sigma^2/2)T$ changes (TN15.5) to $A = S_0 e^{(r-q)T} N(d_1)$ and leaves (TN15.4) in the same form, giving

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}} \tag{TN15.8}$$

Known cash dividends, and the volatility subtlety. Chapter 15 subtracts the present value of dividends and applies the formula to $S^* = S - \mathrm{PV}(D)$. The justification is that S^* is the risky component: S equals a certain part, $\mathrm{PV}(D)$, plus an uncertain part which is assumed lognormal.

The subtlety the chapter flags can be made precise. If S^* has volatility σ^* and $S = S^* + \mathrm{PV}(D)$, then the volatilities of the two are related by

$$\sigma^* \approx \sigma \, \frac{S}{S^*} = \sigma \left(1 - \frac{\mathrm{PV}(D)}{S}\right)^{-1} \tag{TN15.9}$$

because the same absolute risk is carried by a smaller base. For a stock yielding 6% with a two-year option, $\mathrm{PV}(D)/S \approx 0.113$, so σ^* exceeds the observed σ by roughly 13%, which is not negligible. Using the unadjusted volatility understates the option.

A.2.4 Black's Approximation

Chapter 15 states the rule; here is why it takes that form.

Section 11.10 established that an American call on a stock paying a single dividend at t_1 can be optimally exercised only at two moments: immediately before t_1, or at expiration T. Exercise elsewhere is dominated, because the interest saved by deferring the strike payment exceeds any benefit until the dividend is imminent.

The value of an American call is the maximum over admissible exercise policies. Restricting attention to those two:

$$C \approx \max\left(\underbrace{c(S_0 - \mathrm{PV}(D),\, K,\, T)}_{\text{hold to } T},\ \underbrace{c(S_0,\, K,\, t_1)}_{\text{exercise at } t_1} \right) \tag{TN15.10}$$

The first term treats the option as European to T on the dividend-adjusted price. The second treats it as European to t_1 with no adjustment, since a holder exercising then captures the dividend and the adjustment does not apply.

The approximation *understates*, and the reason is now visible: it evaluates a fixed choice between two policies made today, whereas the true American holder decides at t_1 with knowledge of S_{t_1}. Being able to choose after observing is worth something, so

$$C \geq \max(\cdot, \cdot)$$

with equality only in the degenerate cases where one policy dominates in every state. The error is largest when the two terms are close, which is exactly when the exercise decision is genuinely uncertain.

Further Reading

Black and Scholes (1973) derive the formula through a hedged portfolio and the heat equation; Merton (1973) uses a self-financing replicating portfolio and also handles dividends and the American call. Cox and Ross (1976) is where the risk-neutral integral of Section A.2.1 first appears as a general method. For the Feynman–Kac equivalence of Section A.2.2, Baxter and Rennie is the short route and Shreve's second volume the measure-theoretic one.

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